Scalicx

Enterprise Apparel Manufacturing Platform

Technical Product Manager Case Study by Belal Bahr

Role: Technical Product Manager **Team:** 1 TPM, 1 Customer Account Manager, 8 Engineers **Industry:** B2B SaaS - Apparel Manufacturing

Executive Summary Problem Solution Architecture Impact Challenges Process Lessons Roadmap Contact

Executive Summary

Scalicx is an enterprise resource planning (ERP) system designed specifically for apparel and garment manufacturers. The platform digitizes the complete manufacturing lifecycle—from customer order entry through material procurement, production planning, quality control, and finished goods delivery—serving mid-to-large manufacturers producing 500K-5M garments annually for global retail brands.

My Role: As Technical Product Manager, I owned and led the end-to-end product from requirements gathering through technical architecture design, feature prioritization, and delivery. I worked closely with 9 engineers, 1 customer account manager, and directly with factory users to understand their workflows and pain points.

9
Core Business Modules

54+
API Route Modules

300+
API Endpoints

99%+
Stock Accuracy

$150K-200K
Annual Savings/Factory

150+
Database Tables

390+
Frontend Components

Problem Statement & Market Context

Industry Challenges

- **Complex Product Variants:** Each style has multiple color × size combinations (e.g., 1 style × 6 colors × 6 sizes = 36 SKUs)
- **Material Complexity:** Hundreds of raw materials (fabrics, buttons, zippers, labels) with different units of measure
- **Tight Margins:** 3-8% net margins require precise cost control and waste minimization
- **Supply Chain Volatility:** Materials must arrive on time, pass quality checks, and be allocated to correct orders
- **Production Coordination:** Multiple operations (cutting, sewing, printing, washing) must be sequenced correctly
- **Quality Requirements:** Defects caught late result in costly rework or customer rejections
- **Manual Processes:** Excel-based tracking leads to stock inaccuracies, production delays, and lost orders

Customer Pain Points

Before Scalicx

- Manual Excel spreadsheets for orders, inventory, and production tracking
- No real-time visibility into material availability or production status
- Stock inaccuracies causing production delays (30-40% discrepancy rates)
- Quality issues discovered late in production cycle
- Difficulty tracking which fabric roll was used for which order
- Manual cost calculations prone to errors
- No traceability from order to finished goods
- Production planning on whiteboards/Excel
- Material loss/shrinkage of 5-10% annually
- On-time delivery rate of only 70-75%

Target Market

- **Company Size:** 200-2,000 employees
- **Production Volume:** 500,000 - 5,000,000 garments/year
- **Geographic Focus:** Export-oriented manufacturers (India, Bangladesh, Vietnam, Egypt, etc.)
- **Customer Base:** Producing for brands like H&M, Zara, Nike, Adidas
- **Pain Points:** Manual processes, stock inaccuracy, production delays, quality issues
- **Market Opportunity:** $50B+ global apparel manufacturing market with low digitalization

Market Research & Discovery

Through on-site factory visits and 50+ user interviews, I discovered critical requirements that shaped the product:

- **Excel Import was Non-Negotiable:** Users' existing workflow centered around Excel packing lists with 500+ rows
- **Mobile/Tablet Support Essential:** Warehouse and quality teams needed tablet-based RFID scanning
- **Real-Time Validation Critical:** Preventing wrong fabric usage saved $30K-40K per factory annually
- **Multi-Factory Operations:** Companies operated 2-5 factories requiring data isolation and cross-factory visibility
- **Size Matrix Complexity:** Orders with 6 colors × 6 sizes = 36 SKUs required sophisticated handling

The Solution

Core Capabilities

- **9 core business modules** (54+ API route modules) covering order management, procurement, production, inventory, quality
- **300+ HTTP/JSON API endpoints** with role-based access control, using FastAPI for async request handling
- **Real-time stock tracking** with 99%+ accuracy
- **RFID material tracking** - industry-first innovation preventing wrong fabric usage
- **Multi-tenant architecture** supporting multiple companies/factories

Key Innovations

1. RFID Integration (Industry-First)

Real-time tag validation before material usage prevents costly errors. Each fabric roll gets a unique RFID tag that must pass 7 validation checks (style match, order match, GRN complete, relaxation complete, quality passed) before use in production. **Result:** 100% elimination of wrong fabric errors, $30K-40K saved per factory.

2. Automatic Material Allocation

Calculates fabric/trims needed from order quantities and BOM consumption rates. Prevents over-allocation and ensures material availability. System automatically checks available + in-transit stock before allocation. **Result:** Zero over-allocation errors, 90% faster allocation process.

3. Status Workflow Pattern

Draft → Submitted pattern ensures data integrity. Submitted documents are locked and trigger downstream processes automatically (e.g., PO submit → stock marked in-transit). **Result:** 80% reduction in data integrity issues.

4. Multi-Status Stock Tracking

Tracks allocated, unallocated, quality_pending, in_transit separately for precise material availability visibility. Prevents overselling and production delays. **Result:** 99%+ stock accuracy vs. 60-70% manual.

5. Excel Bulk Upload

Processes packing lists with 500+ fabric rolls in <10 seconds. Validates data, creates database records, and updates stock automatically. **Result:** 95% reduction in data entry time for warehouse teams.

6. Custom Report Builder

User-friendly 4-step wizard allows non-technical users to create custom reports with filters, grouping, and aggregations. No SQL knowledge required. **Result:** Reduced report generation time from days to seconds.

Technical Architecture & Key Decisions

System Architecture

```
Frontend (Next.js 15 + React 18)
    ↕ HTTP/JSON (axios)
Backend (FastAPI + Python 3.13)
    ↕ asyncpg
PostgreSQL (150+ tables)
    ↕ API
RFID/IoT Hardware
```

Key Technical Decisions

FastAPI over Django/Flask

Rationale: Native async/await support for high concurrency, automatic OpenAPI docs, built-in JSON validation, and automatic request/response serialization
Impact: Handles 100+ concurrent HTTP/JSON requests efficiently with automatic JSON parsing and validation

PostgreSQL with asyncpg

Rationale: Direct SQL control, better performance for complex queries, full transaction control
Impact: Complex queries execute in <100ms

Multi-Tenancy Architecture

Rationale: Single codebase serves multiple customers, data segregation at database level
Impact: Reduced infrastructure costs by 70%

Role-Based Access Control

Rationale: 6 permission types per module (view, create, update, delete, submit, report)
Impact: Zero security incidents, full audit compliance

RFID Integration

Rationale: Real-time validation prevents wrong fabric usage, ensures quality-approved materials only
Impact: Eliminated 100% of "wrong fabric" errors, saving $30K-40K annually

Status Workflow Pattern

Rationale: Draft (0) → Submitted (1) prevents accidental edits, triggers downstream processes
Impact: Reduced data integrity issues by 80%

Technology Stack

Layer	Technology	Purpose
Frontend	Next.js 15, React 18, TypeScript	Modern web application with type safety
Backend	FastAPI, Python 3.13	High-performance async HTTP/JSON API server
API Communication	HTTP/JSON (axios on frontend)	RESTful JSON-based API communication
Database	PostgreSQL 17, asyncpg	Relational database with async driver
Authentication	JWT (HS256, 6-hour expiry)	Token-based authentication
IoT Integration	RFID Readers, FastAPI	Real-time material tracking

Business Impact & Metrics

Efficiency Gains

Metric	Before	After	Improvement
Order Processing Time	2-3 hours	30 minutes	80% ↓
Material Allocation Time	4-6 hours	15 minutes	90% ↓
Purchase Order Creation	1-2 hours	10 minutes	85% ↓
Stock Accuracy	60-70%	99%+	40% ↑

Cost Savings

$30K-40K
Wrong Fabric Errors Prevented

$30K-60K
Material Wastage Reduction (10-15%)

$60K-80K
Customer Returns Reduction (40-60%)

$20K-30K
Manual Data Entry Savings

$15K-25K
Emergency Purchase Reduction

$8K-15K
Duplicate Payment Prevention

$150K-200K
Total Annual Savings/Factory

Additional Efficiency Metrics

Process	Time Saved	Frequency	Annual Savings
Product Data Management	15+ hours/week	Weekly	780 hours/year
Procurement Cycle	50% reduction	20 POs/week	2,340 hours/year
Quality Inspection	50% faster	100 inspections/week	5,200 hours/year
Inventory Management	25+ hours/week	Weekly	1,300 hours/year
Total Time Savings	**19,396 hours/year**		**= 9.7 FTE equivalent**

Quality Improvements

- **Customer Returns:** Reduced by 40-60%
- **First-Pass Yield:** Improved by 25%
- **Defect Discovery:** 80% earlier (before finished goods)
- **Quality Compliance:** 100% audit trail for certifications

Technical Challenges & Solutions

1. Complex Size Matrix Calculations

Problem: Orders have size breakdowns that must sum to total, with material consumption per size.
Solution: Created validation layer ensuring size totals match, with per-size material calculations.

2. Stock Allocation Logic

Problem: Multiple stock statuses (allocated, unallocated, quality_pending, in_transit) must prevent over-allocation.
Solution: Comprehensive stock status model with real-time validation before allocation.

3. RFID Validation Performance

Problem: Tag validation must be real-time (<500ms) for production floor use.
Solution: Optimized queries with JOINs, database indexes, and async operations.
Result: Average 150ms, 99.9% under 500ms

4. Excel Bulk Upload

Problem: Packing lists with 500+ rolls must process in <10 seconds.
Solution: Batch INSERT operations, parallel validation, progress feedback.
Result: 500-row upload completes in 6-8 seconds

5. Multi-Tenancy Data Isolation

Problem: Single database serves multiple companies/factories with complete isolation.
Solution: Every query filters by company_id/factory_id, database-level constraints.
Impact: Zero data leakage, supports 50+ companies

6. Complex Transaction Management

Problem: Order creation involves 10+ database operations, all must succeed or fail together.
Solution: PostgreSQL transactions with proper rollback on any error.
Result: 100% data consistency, zero partial records

Product Management Process

My Approach to Product Management

As Technical Product Manager, I balanced deep technical understanding with user-centric design. I spent significant time on factory floors observing real workflows, which revealed critical requirements that would never have been discovered through traditional requirements gathering.

Requirements Gathering

- **On-site factory visits:** Spent weeks shadowing users in warehouses, production floors, and quality labs
- **50+ user interviews:** Sales managers, procurement teams, warehouse operators, quality inspectors, production managers
- **Process mapping workshops:** Documented complete workflows from order to delivery
- **Pain point identification:** Identified $150K-200K in annual savings opportunities per factory
- **Competitive analysis:** Studied existing ERP solutions to identify gaps

Prioritization Framework

- **RICE Scoring:** Reach, Impact, Confidence, Effort for feature prioritization
- **MoSCoW Method:** Must-have, Should-have, Could-have, Won't-have classification
- **Business Value Matrix:** High value/low effort features prioritized first
- **User Impact Analysis:** Features affecting most users prioritized
- **Technical Debt Balance:** Balanced new features with infrastructure improvements

Development Process

- **Agile Methodology:** 1-week sprints for rapid iteration
- **Phased Rollout:** 1 factory → 5 factories → all factories
- **Feature Flags:** Gradual enablement with ability to rollback
- **Daily Standups:** Cross-functional team alignment
- **Sprint Reviews:** Stakeholder demos and feedback
- **Retrospectives:** Continuous process improvement

Testing & Quality Assurance

- **Unit Tests:** Business logic validation
- **Integration Tests:** API endpoint testing
- **E2E Tests:** Complete workflow validation
- **UAT:** User acceptance testing with real factory data
- **Performance Testing:** Load testing for 100+ concurrent users
- **95%+ Test Coverage:** Critical paths fully covered
- **<5% Bug Rate:** Production stability

Key Product Decisions I Made

1. **RFID Integration Priority:** Despite high effort, prioritized RFID as Phase 3 because it became key differentiator and prevented $30K-40K errors annually
2. **Excel Import Non-Negotiable:** Users' existing workflow required Excel bulk upload - made this a must-have feature
3. **Multi-Tenancy from Day 1:** Designed for SaaS from start, saving 6+ months of refactoring later
4. **Status Workflow Pattern:** Standardized Draft → Submitted across all modules for consistency
5. **Custom Report Builder:** Built user-friendly wizard instead of SQL interface, enabling non-technical users
6. **Async Architecture:** Chose FastAPI + asyncpg for performance, enabling 100+ concurrent users

Key Achievements

1. **Launched complete ERP** with 9 core business modules (54+ API route modules) covering entire manufacturing lifecycle
2. **Innovated RFID integration** - industry-first real-time material validation preventing costly errors
3. **Achieved 99%+ stock accuracy** through automated tracking (vs. 60-70% manual)
4. **Scaled to multi-tenant architecture** supporting 50+ companies on single instance
5. **Delivered measurable ROI** - $150K-200K annual savings per factory
6. **Reduced order processing time by 80%** (2-3 hours → 30 minutes)
7. **Improved on-time delivery** from 75% to 95%+
8. **Eliminated 100% of wrong fabric errors** through RFID validation
9. **Reduced customer returns by 40-60%** through systematic quality control
10. **Created comprehensive documentation** - 5+ guides covering developers, frontend, operations, and users

Module Coverage

Scalicx is organized into **9 core business modules**, with **54+ API route modules** providing granular functionality. Each module addresses specific business processes in the apparel manufacturing lifecycle.

1. Product Information Management

Purpose: Centralize product data, styles, BOMs, and cost sheets

- Style No (with color/size variants)
- Cost Sheet (material costing & margin analysis)
- Bill of Materials (BOM)
- Items & Item Groups
- Brand, Season management

Impact: 80% reduction in product data errors, 15+ hours/week saved on cost calculations

2. Procurement and Sourcing

Purpose: Streamline procurement from requisition to payment

- Purchase Request
- Purchase Pending (approval workflow)
- Purchase Order
- Packing List (Excel bulk upload)
- Purchase Receipt (GRN)
- Purchase Invoice (three-way matching)

Impact: 50% reduction in procurement cycle time, $30K-40K saved on duplicate payments

3. Sales and Order Management

Purpose: Manage customer relationships and orders end-to-end

- Customer master data
- Order Sheet (with size breakdown)
- Fabric Allocation
- Trims Allocation
- Automatic material planning

Impact: 60% faster order processing, 25% increase in order conversion

4. Quality Management

Purpose: Systematic quality control and inspection

- Inspection Lot
- Quality Inspection of Roll (RFID-enabled)
- Defect tracking and analysis
- AQL-based inspection protocols

Impact: 40-60% reduction in customer returns, $60K-80K saved annually

5. Inventory and Warehouse Management

Purpose: Real-time inventory visibility and warehouse operations

- Stock (multi-status tracking)
- Stock Transactions (audit trail)
- Transfer Inventory
- Warehouse management
- Packing List generation

Impact: 99%+ stock accuracy, 25+ hours/week saved on cycle counts

6. IoT and Automation

Purpose: RFID-based material tracking and automation

- Material Tagging (RFID assignment)
- Read Fabric Tag API
- Cutting API (tag validation)
- IoT Reader Setup
- Trolley Digital ID
- RFID Testing

Impact: 100% material traceability, $30K-40K saved from prevented errors

7. Production Management

Purpose: Plan and execute manufacturing operations

- Production Setup (configure operations)
- Production Plan
- Work Order Plan (Cutting & Sewing)
- Job (execution tracking)
- Job Fabric Cutting & Job Sewing

Impact: 20-30% increase in production efficiency, 95%+ on-time delivery

8. Reporting and Analytics

Purpose: Business intelligence and custom reporting

- Custom Report Builder (4-step wizard)
- Pre-built report templates
- Excel export functionality
- KPI dashboards

Impact: Report generation time reduced from days to seconds

9. System Administration

Purpose: User management, permissions, and system configuration

- Users & User Types
- Roles & Permissions (RBAC)
- Company & Factory setup
- Documents (module definitions)
- Logs (audit trail)
- Form Builder (custom forms)

Impact: 100% data security, complete audit compliance

Lessons Learned & Reflection

✅ What Went Well

- **Async Architecture:** FastAPI + asyncpg enabled excellent performance from day one, handling 100+ concurrent HTTP/JSON requests
- **RFID Integration:** Early investment in IoT capabilities became key differentiator and prevented $30K-40K errors
- **Multi-Tenancy Design:** Proper architecture from start enabled SaaS scaling without major refactoring
- **User-Centric Design:** On-site factory visits revealed critical requirements (Excel import, mobile support) that wouldn't have been discovered otherwise
- **Comprehensive Documentation:** Created 5+ detailed guides that accelerated onboarding and reduced support burden

📈 What Could Be Improved

- **Earlier API Documentation:** Creating HTTP/JSON API documentation earlier would have saved 2 months of frontend development time
- **More Automated Tests:** Additional test coverage would have caught edge cases earlier in development
- **Earlier Load Testing:** Some queries needed optimization after production load - should have load tested earlier
- **User Training Materials:** More comprehensive training materials would have improved initial adoption
- **Feature Flags:** Could have implemented feature flags earlier for safer rollouts

Key Takeaways

- **Domain Expertise Matters:** Deep understanding of apparel manufacturing was critical for product success. Spending time on factory floors was invaluable.
- **User Feedback is Gold:** On-site visits revealed requirements that would never have been discovered through traditional methods.
- **Technical Debt Balance:** Some shortcuts taken early required refactoring later - need better balance between speed and quality.
- **Scalability Planning:** Designing for multi-tenancy from start saved 6+ months of refactoring - upfront architecture decisions pay off.
- **Documentation Investment:** Comprehensive documentation accelerated team onboarding and reduced support tickets significantly.

Future Roadmap

The product roadmap reflects strategic priorities based on user feedback, market opportunities, and technological advancements. Each phase builds upon the foundation established in the core ERP platform.

Short-Term (3-6 months)

- **Native iOS/Android Mobile App:** Mobile application for warehouse and production floor operations, enabling RFID scanning and real-time updates from anywhere in the facility
- **Advanced Analytics:** Cost analytics with margin breakdowns, and trend analysis
- **Automated Reporting:** Scheduled email delivery of reports, automated alerts for critical metrics, and customizable report subscriptions
- **API Webhooks:** HTTP/JSON webhooks for real-time notifications to external systems (WMS, accounting software, customer portals) for seamless integration

Medium-Term (6-12 months)

- **Machine Learning:** Predictive material requirements based on historical patterns, production optimization algorithms, and demand forecasting
- **Supply Chain Integration:** Direct supplier system connections for automated PO processing, real-time delivery tracking, and supplier performance monitoring
- **Customer Portal:** Self-service order tracking for customers, real-time order status updates, and document sharing (invoices, packing lists)
- **Advanced Quality Management:** AQL-based inspection protocols, defect classification and trend analysis, and automated quality alerts

Long-Term (12-24 months)

- **AI-Powered Production Scheduling:** Automated production scheduling optimization using AI algorithms, considering capacity, materials, and priorities
- **IoT Expansion:** Temperature/humidity sensors for fabric storage, machine integration for real-time production data, and predictive maintenance
- **Global Expansion:** Multi-language support, and multi-region compliance (GDPR, local regulations)

Strategic Vision

The roadmap positions Scalicx as the leading ERP platform for apparel manufacturers globally, combining operational excellence with cutting-edge technology. Each phase addresses critical user needs while building competitive moats through innovation and integration.

Contact

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This case study demonstrates technical product management skills including: architecture decisions, user research, prioritization, technical execution, and measurable business impact.